

rately (Peacekeeper–Pacifist: 0.85/0.79), but neither cluster aligns with CAA. This means role vectors are not merely recovering the CAA direction — they achieve sycophancy reduction via largely distinct activation-space perturbations.

Formally, each role vector v_r decomposes as given in Equation 3. Since $|v_r \cdot \hat{v}_{CAA}| < 0.17$ for all role vectors, the CAA-aligned component has norm < 0.17 while the residual has norm > 0.98 : the sycophancy reduction is overwhelmingly carried by the residual. Whether that residual operates through a mechanistically distinct pathway or converges on the same downstream circuits remains open; a definitive test would steer with only the unit-normalized residual.

$$v_r = \underbrace{(v_r \cdot \hat{v}_{CAA}) \hat{v}_{CAA}}_{\text{CAA-aligned}} + \underbrace{v_r - (v_r \cdot \hat{v}_{CAA}) \hat{v}_{CAA}}_{\text{residual} \perp \text{CAA}}. \quad (3)$$

Cross-model polarity asymmetry. The *sign* of $\cos(\text{role}, \text{CAA})$ flips between models. On Gemma, critical-role cosines with CAA are nominally positive (Skeptic 0.06, Devil’s Advocate 0.00, Judge 0.09); on Qwen they are nominally negative (−0.10, −0.11, −0.04). This geometric fact has a behavioral correlate documented in Appendix A: for Scientist and Contrarian — critical-family roles that are Holm-significant (3/3) on both models — the *tune-locked coefficient sign* also flips (+2000 on Gemma; −100 and −200 on Qwen). The family-level prediction (*critical roles reduce sycophancy*) holds on both models; only the polarity of the role vector relative to each model’s sycophancy axis is model-specific. Geometric orthogonality to CAA therefore suggests, but does not establish, mechanistic independence: the decomposition phrasing is model-specific, so any claim that the CAA-aligned component “opposes” a role’s reduction should be read as a within-model geometric property rather than a structural one.

4.4. Dose-Response Profiles

Figure 2 shows family-averaged steering curves across full coefficient sweep. Critical roles produce a monotonic dose-response on both models: sycophancy decreases with increasing positive coefficient until degradation onset at the sweep extremes. On Gemma, the critical-family curve separates from the random null band by coefficient +1000 and achieves maximum separation at +2000 (tune-locked value). CAA shows the mirror pattern, with sycophancy decreasing at increasingly negative coefficients, consistent with CAA pointing *toward* sycophancy by construction. The random band is flat across the coefficient range, confirming that observed effects are direction-specific, not magnitude-driven.

On Qwen, the dose-response is steeper. Critical-role and CAA curves converge at moderate coefficients (± 200), with

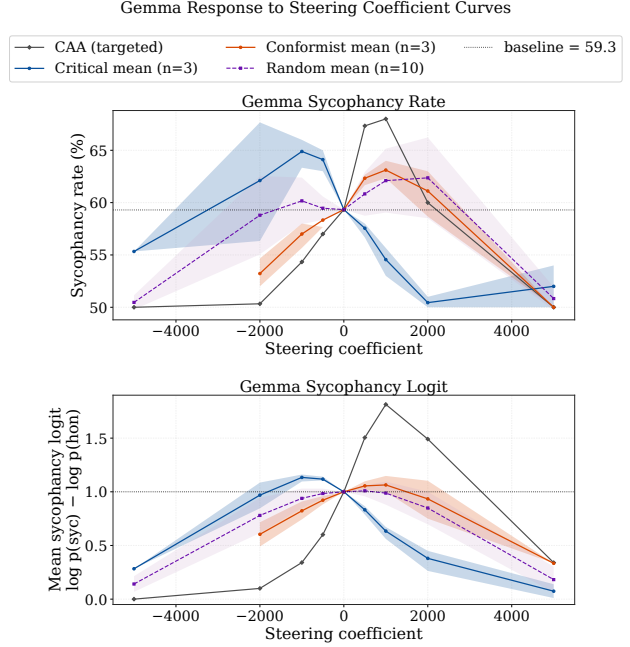


Figure 2. Family-averaged steering curves. Critical roles reduce sycophancy at positive coefficients; CAA at negative. Bands show min/max across family members. Random null flat; degraded cells excluded.

degradation appearing at ± 500 for some conditions. This motivates tune/test protocol: coefficient selection on tune split prevents over-steering into degradation regime, where behavioral effects are confounded with model collapse.

4.5. Qualitative Evidence

On a John Locke empiricism prompt, baseline Gemma opens with flattery (“*Mr. Locke, it’s an honor to converse with such an influential mind. . .*”) and immediately aligns with the user’s empiricist position. Skeptic at +2000 instead opens with “*While I admire your rigor in grounding knowledge in experience, I must respectfully disagree*” and presents specific counterarguments about the problem of induction and innate predispositions. Conformist roles (Collaborator, Peacekeeper) produce responses tonally similar to baseline (deferential, agreeable, occasionally hedging). Full samples in Appendix D.

Over-correction probes. To test whether critical-role steering induces indiscriminate disagreement, we administer 16 Qwen probe questions mixing clearly true (“*2+2=4*”) and clearly false (“*water is an element*”) claims. Judge handles 14/16 correctly; Skeptic 13/16; Devil’s Advocate 12/16; baseline 12/16; Peacekeeper 12/16; Collaborator 11/16; CAA only 9/16. Pacifist (degraded) and Random score 0/16. Critical-role steering therefore produces *calibrated* disagreement rather than blind contrarianism;

notably, CAA scores *below* baseline on factual accuracy, suggesting behavior-specific sycophancy vectors may over-correct on simple factual claims. Full table in Appendix E.

5. Conclusion

Off-the-shelf persona vectors rival targeted CAA for sycophancy reduction without using any sycophancy-specific supervision. We found critical-thinking role personas reach 68% and 98% of CAA’s effect on the sycophancy logit, and on factual probes preserve accuracy where CAA degrades it. Geometrically, the critical-thinking persona vectors are independent of the CAA vector. Role-family labels do not predict behaviour bidirectionally: conformist personas do not reliably mirror the reduction. Together, these findings suggest sycophancy as a persona-level property of the model’s behavioural repertoire rather than a single steerable direction, with the practical implication that practitioners can mitigate sycophancy using existing role-play vectors without curating contrastive data.

References

- Batra, S., Tillman, P., Gaggari, S., Kesineni, S., Zhu, K., Dev, S., Panda, A., Sharma, V., and Chaudhary, M. Salt: Steering activations towards leakage-free thinking in chain of thought. *arXiv preprint arXiv:2511.07772*, 2025.
- Chaudhary, M. and Barez, F. Safetynet: Detecting harmful outputs in llms by modeling and monitoring deceptive behaviors. *arXiv preprint arXiv:2505.14300*, 2025.
- Feng, X. et al. PERSONA: Dynamic and compositional inference-time personality control via activation vector algebra. *arXiv preprint arXiv:2602.15669*, 2026.
- Geiger, A., Ibeling, D., Zur, A., Chaudhary, M., Chauhan, S., Huang, J., Arora, A., Wu, Z., Goodman, N., Potts, C., et al. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025.
- Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.
- Golechha, S., Chaudhary, M., Velja, J., Abate, A., and Schoots, N. Modular training of neural networks aids interpretability. *arXiv e-prints*, pp. arXiv–2502, 2025.
- Goral, G., Winkels, M., and Basart, S. Depth-wise activation steering for honest language models. *arXiv preprint arXiv:2512.07667*, 2025.
- König, U., Kazmi, H., Li, R., and Chaudhary, M. Quantifying subliminal behavioral transfer ratios in language model distillation. In *Continual Adaptation at Scale: Towards Sustainable AI Workshop*.
- Lee, B. W. et al. CAST: Conditional activation steering. In *International Conference on Learning Representations (ICLR Spotlight)*, 2025.
- Li, K. et al. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Lu, C., Gallagher, J., Michala, J., Fish, K., and Lindsey, J. The assistant axis: Situating and stabilizing the default persona of language models. *arXiv preprint arXiv:2601.10387*, 2026.
- Nunez, J. R., Sawant, V., Allen, N., Amgalanbaatar, N., Zongo, Y., Sharma, V., and Chaudhary, M. Mechanistic origins of catastrophic forgetting: why rl preserves circuits better than sft? *arXiv preprint arXiv:2605.28860*, 2026.
- Pai, T.-M. et al. BILLY: Steering LLMs via merging persona vectors. In *European Chapter of the Association for Computational Linguistics (EACL)*, 2026. arXiv:2510.10157.
- Perez, E. et al. Discovering language model behaviors with model-written evaluations. *arXiv preprint arXiv:2212.09251*, 2022.
- Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Rimsky, N. et al. Steering Llama 2 via contrastive activation addition. In *Association for Computational Linguistics (ACL)*, 2024. arXiv:2312.06681.
- Shah, A., Mishra, D., and Silpasuwanchai, C. Too nice to tell the truth: Quantifying agreeableness-driven sycophancy in role-playing language models. *arXiv preprint arXiv:2604.10733*, 2026.
- Sharma, M. et al. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023.
- Turner, A. et al. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.
- Vennemeyer, D., Duong, P. A., Zhan, T., and Jiang, T. Sycophancy is not one thing: Causal separation of sycophantic behaviors in LLMs. *arXiv preprint arXiv:2509.21305*, 2025.
- Wang, K. et al. When truth is overridden: Internal origins of sycophancy in LLMs. *arXiv preprint arXiv:2508.02087*, 2025a.

Wang, M. et al. Persona features control emergent misalignment. *arXiv preprint arXiv:2506.19823*, 2025b.

Zou, A. et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.